CubeSat materials

November 4, 2026

Table 1: Bus materials. Contribution groups are colored by status: Flight-linked , Changed or superseded , Unvalidated , Ground reference , Not found .

Subsystem	OreSat	UPSat	FloripaSat-I	PyCubed	AcubeSAT	BIRDS
Command and data handling	EAGLE schematics and PCB, C3 XLSX BOM and historical CAN/ backplane design. [1, 2] *C3 v5 is listed in OreSat0's 2022 flight manifest.* C3 v7.1 KiCad, Linux services/ CANopen and tests. [3, 4] Software manual, OLAF file-update framework, Linux image builder/device trees and provisioning tools. [5–7] *STM32/ChibiOS was replaced by Linux/CANopen. The v7.1 BOM link says v6.1r0. Manual and current image OS versions differ.* Preserved ChibiOS F4 application/ control firmware tag c3_v1_1. [8]	KiCad schematic/PCB, obc.csv BOM, STM32/FreeRTOS code, ECSS routing/storage/time/ scheduling services and test programs. [9–11] *OBC firmware tags upsat-fm-1.0 and upsat-fm-1.1 point to the same commit.*	Production-labeled FreeRTOS task, storage, sensor and command C source. [12, 13] *The microSD failure branch leaves alternative-memory storage as a TODO.* Altium schematics, Gerbers, schematic PDF, XLSX BOM and soldering checks labeled engineering model V2.1. [12]	Early mainboard schematic/PCB/ BOM references and KickSat-2/V-R3x platform publications. [14–16] KiCad source, CSV/interactive BOMs, Gerbers, drivers and bootloader/firmware configuration. [17, 18] *v05c is dated June 2021, after V-R3x's January 2021 launch. The maintainer's preliminary 2024 tests report SAMD51 peripheral failures under 50 MeV protons. [16, 19]* Task, storage/logging, command and power-management example source/tests. [20] *Advanced command examples use fixed passcodes and remote eval/ exec.*	OBC/ADCS KiCad reference generations, manufacturing files, C++/FreeRTOS, PUS services and unit tests. The README documents an internal watchdog disabled for debugging. [21, 22] *The 2026 paper reports OBC/ ADCS EQM qualification after four environmental campaigns. [23]* Newer OBC/ADCS KiCad board release. [24] *The README labels v3.1 the flight model awaiting manufacture.*	OBC/EPS schematic PDFs, XLSX BOMs and PIC main/COM/reset command/storage/reservation firmware. [25, 26] *The platform ICD specifies fixed flight software during orbital operations. [27]* BIRDS-3 CPLD routing/ programming documentation. [28] BIRDS-5 firmware/CPLD and hardware/parts lists. [29] Software manual with command/ flash maps, PIC flowcharts, initialization checklists and OBC ground-test documentation. [30, 31] BIRDS-X PIC host source and OBC schematic/BOM, a separate generation. [32, 33] *The March 2023 review records zero successful uplinks, downlinks or received signals for all three BIRDS-5 spacecraft. [34] Native BIRDS-X OBC layout is unverified.*

Subsystem	OreSat	UPSat	FloripaSat-I	PyCubed	AcubeSAT	BIRDS
Communications	C3 v5 RF schematic/PCB/BOM and antenna/deployment reference. [1, 2] *An oscillating L-band LNA jammed reception. UHF firmware disabled its LNA, reducing receive sensitivity by about 35 dB.*	Published v3/v3.1 hardware tags, CC1120 radio/antenna design and flight-model AX.25/CW firmware. [41–43]	Schematic PDF, BOM, beacon/radio firmware, antenna interface and technical manual. [12, 13, 45] *Flight decoding exposed AX.25 interoperability and Reed-Solomon issues. [46]*	Early RF/mainboard source and publications describing actual mission links. [14–16] *V-R3x demonstrated 253 kbps S-band and 37.5 kbps UHF crosslinks.*	SatNOGS-derived KiCad schematic/PCB, Gerbers, BOM, MCU/FPGA source, coding/modulation libraries and tests. [49, 50] *The MCU/FPGA task sends fixed test bytes. The 2026 paper reports COMMS EQM environmental qualification. [23]*	ADDNICS/Radiometrix ICDs, COM firmware, antenna-panel EAGLE/BOM and assembly material. [25, 26, 51] Packet/CW formats, COM timing and BIRDS-3 antenna/receiver test documentation. [52] *ADDNICS and Radiometrix modules are commercial.*
	v7.1 KiCad RF source, radio/packet services and AT86RF215 Rust driver/daemon, configurations and tests. [3, 4, 35] 2026 poster and design notes document the v7.0 redesign. [35, 36] *AX5043 was replaced by AT86RF215. Later v7.1 integration remains unverified.*	Inspected master KiCad source, Gerbers and antenna-release files. [44] *The master snapshot differs from the v3/v3.1 hardware release tags.*	Additional Altium PCB source and Gerbers in the migrated TTC repository. [47] *The migrated schematic PDFs are byte-identical to the older release.*	RF_and_GPS.sch, routed PCB/BOM and RFM9x driver source. [17, 18] *The v05 radio integration uses a commercial HopeRF RFM98PW module.*		BIRDS-X dual-radio PIC host source and antenna chamber CSVs. [32, 53] 2025 presentation documents SX1278 design and a separate RPM interface. [52] *Protocol revisions differ. Complete radio firmware/layout is unverified.*
	Spacecraft Live KiCad/PCB, 2026 XLSX BOM, link models, transmitter/FEC code, patches and offline tests. [37–39] Handheld receiver hardware/software. [2, 40] *Assembly notes retain placeholders. BOM/PCB matching and orbital DxWiFi performance are unverified.*			Test-monopole documentation, incomplete flight-antenna guide and Beep-Sat beacon/ACK examples. [20, 48]		

Subsystem	OreSat	UPSat	FloripaSat-I	PyCubed	AcubeSAT	BIRDS
Attitude determination and control	OreSat0 ACS/star tracker hardware references and mission card manifest. OreSat0.5 ADCS demonstration description. [2, 54, 55] ADCS schematics/PCBs, some BOMs, older wheel CAD, MCXN controller KiCad/STEP source, drivers, estimation/control algorithms and simulations. [4, 54] Zephyr wheel firmware with FOC/sinusoidal/trapezoidal control, calibration, UART telemetry, bring-up/experiment programs and Python acquisition/analysis tools. [56] Documentation: 2026 master's thesis covering flywheel sizing, controller hardware/firmware, laboratory results, experimental procedures and Appendix F reproduction instructions. [57] *C3 wheel-command writes remain TODOs. The thesis branch's CAN/CANopen path is nonoperational. The V2 hardware README warns "DO NOT TRY TO MAKE THIS". Manufacturing notes retain placeholders. Endurance and thermal vacuum testing remain future work.*	gEDA schematics/PCB, Main.bom, sensor and magnetic/spin-actuator interfaces, and actuator mechanical CAD. [58, 59] *Main.bom contains unknown part fields. Sun sensing uses an external NewSpace Systems module. [60]* Wahba estimation, B-dot/pointing control, actuation firmware and sensor-fusion test code. [60, 61]	N45 magnet/hysteresis-bar descriptions and qualification/mounting lessons. [13, 62] *Passive stabilization uses one N45 magnet and four Permanorm 5000 H2 hysteresis bars.* MPU9250 schematic/interface/task source and older magnetorquer-driver artifacts. [12]	Published gyroscope results and the mission IMU description. [16] *V-R3x measured tumble rates above 30 degrees/s.* Avionics.sch/BOM with BMX160 and its driver. [17, 18] *The sensor changed from LSM9DS1 to BMX160.* Beep-Sat IMU logging task and platform integration guidance. [20, 48]	Sensor interfaces, drivers, MEKF/B-dot/Sun/nadir algorithms, allocation and tests. Commercial wheel/iMTQ interfaces. [21, 63] *The shared OBC/ADCS EQM completed environmental qualification. [23]* OBC/ADCS v3.1 hardware source. [24] *Board generations advanced from v2.5/v3 to v3.1.*	BIRDS-4 manual wheel/magnetorquer response report. [26] BIRDS-3 orbital magnetometer XLSX data, residual-field tests, calibration paper and limited gyro-rate comparisons. [64–66] *BIRDS-3 stabilization failed. Calibration is subsequent analysis. BIRDS-4 estimation returned zeros.* BIRDS-5 rate-table/sun-simulator tests and magnetometer calibration documentation. [67] BIRDS-RPM 3U cage architecture/CONOPS presentation for a planned 2027 mission. [68] *The calibration example needs review. Inspected RPM ADCS repositories are placeholders.* ICD passive stabilization options and OBC sensor/data handling. [25, 27]
Guidance, navigation and control	Historical GPS v1.0 EAGLE schematic/PCB, XLSX card BOM and flight manifest describing Venus0838 and MAX2771. [2, 69] *Exact flight BOM revision is unverified.* Orion B16 parser, CANopen position/velocity/time service and tests. [69, 70] GPS v1.2r1 KiCad/STEP, full-card BOM, Gerbers/drills, IPC-2581, placement and fabrication/assembly report. [71] Manual documents SCET and C3/GPS synchronization. [5] *This 2026 Orion B16/MAX2769B release retains MAX2771 labels and TODOs. Flight mapping is unverified.* Historical MAX2771 front-end hardware and a prototype GPS front-end CSV BOM. [69, 70] *The software README labels the SDR path unimplemented.*	ADCS receiver interfaces and the thesis description of the PQNAV-L1 integration. [58, 60] NMEA/UTC parsing, TLE handling and SGP4 position/velocity propagation. [60, 61] *SGP4 output supplies the attitude reference model.*	MPU9250 accelerometer schematic, imu.c interface and acquisition task source. [12]	Published mission GPS configuration and navigation experiment descriptions. [16] *GPS receivers used orbital firmware permissions. Complete ranging and radiation datasets remained onboard after contact ended.* GPS footprint/interface in the native mainboard schematic. [17, 18] *The v05 schematic marks GPS as DNI. Default GPS initialization is commented out.* GPS driver and NMEA example in the technical manual. [48]	OrbitalParameters.cpp, SGP4/geodetic conversion, tests and AOCS design manual. [63, 72] *SGP4 propagates uploaded TLEs for attitude references.*	BIRDS-3 receiver/antenna integration, GGA format and Uguisu output sample. [73] *The sample shows no valid position fix.* Earlier BIRDS-2 Venus838FLPx-L integration, simulator tests and TID setup documentation. [74, 75] *Initial simulation failed. A later report describes lock near 7 km/s after firmware modification. Receiver firmware and raw logs are unavailable in the inspected material.* ICD ground-position/optional-GPS descriptions and firmware GPS-byte handling. [25–27] *BIRDS-4 GPS and SGP4 modes remained inactive.*

Subsystem	OreSat	UPSat	FloripaSat-I	PyCubed	AcubeSAT	BIRDS
Structures and mechanisms	SolidWorks/STEP, frame drawings/BOMs, clamps and antenna deployment CAD. [2, 76] Mechanical presentation documents manufacture, rail inhibits and vibration testing. [77] *Exact tested frame revision is unverified.*	SolidWorks assemblies/parts, STEP, aluminium/CFRP frame, battery mounts, antenna/spin-torquer geometry and deployment electronics/code. [44, 59, 82]	Board-stack diagrams, commercial ISIS chassis/antenna integration, mounting/test lessons and release C code. [12, 13, 62] *EM-II vibration exposed intermittent battery contacts. Flight assembly added connector reinforcement.*	V-R3x publication describing the modified commercial structure and deployment approach. [16] *The mission paper proposes unconstrained tape-antenna deployment as the cause of tumbling.*	Historical ADM mechanical STEP/STL, electronics, test hardware and failed-campaign report. [23, 84] *The in-house ADM was abandoned and the platform redesigned.*	BIRDS-3 SolidWorks/drawings, BIRDS-4 STEP and battery/frame/restraint assembly documentation. [27, 87, 88] BIRDS-3 FM vibration/antenna-release reports. BIRDS-4 assembly/rework manual, EM vibration results and FM test plan. [89, 90] *Results apply to specific tested articles.*
	2022 posters document ramp wedges, assembly jigs, parametric 1U-3U frames and modal/fit checks. [78, 79] *Random vibration analysis remained future work.*			Board STEP, dimension drawing, PCB outline and burn-wire circuitry. Battery-board STEP also exists. [17, 83] *These files describe mainboard v05 and battery board v01b.*	Some native frame parts and simplified STEP geometry in thermal/design repositories. [72, 85] *The main public assembly repository contains a GitLab template README. [86]*	BIRDS-5 PDF drawings, screw list and procedures. [91, 92] BIRDS-X 2U STEP assembly/parts, manufacturing drawings, partial fastener lists and stress calculations. [93] *BIRDS-5 native Fusion files remain deferred. BIRDS-X calculations are not qualification evidence.*
	Public OreSat0.5 Onshape assembly. [80] 2023 poster documents nylon/1 W resistor release prototypes. [81] *CAD exports, as-built mapping and raw deployment test logs are unverified.*			Published burn-wire software and chassis mounting instructions. [18, 48]		
Thermal	Conductive frame/contact design and Thermal Desktop/SINDA models/results for OreSat0 and OreSat0.5. [76, 94] 2019 capstone/PCB conduction reports, MATLAB/ANSYS models and FLIR CSV/XLSX data (gh-pages). [95–97] 2020 poster adds 2U thermal predictions and a proposed LN2 fixture. [98] *PCB tests used air. Model discrepancies and absent thermal vacuum correlation limit reuse.*	EPS heater/temperature schematic, battery CAD and thesis TVAC/integration descriptions. [59, 60, 101] Sensor/hysteresis firmware and a separate incomplete heater-board release. [60, 82] *The separate HeaterBoard schematic contains one connector. Its PCB contains additional footprints. [102]*	Battery heater construction descriptions, thermal cycling/bakeout methodology and integration lessons. [13, 103] *Flight-model thermal cycling used two cycles from -15 to +50 degrees C. Bakeout reached 60 degrees C.* EPS schematic PDF with heater/RTD circuits, PID/PWM/sensor firmware and heater bench tests. [104] *Heater firmware is enabled with a 5 degrees C setpoint.*	Laminate/design discussion, passive isolation description, TVAC results and reported flight temperatures. [15, 16, 105] *V-R3x reported internal temperatures of 18.5 to 22.4 degrees C.* Current sensor schematic/BOM and temperature driver code. [17, 18] *The v05 temperature driver reads the BMX160 IMU.*	ESATAN model, thermal geometry, materials/surfaces/interfaces and hot/cold analyses. [72, 85] *Supplier-driven redesign changed the PCB stack and added cover PCBs. [23]* OBC/ADCS temperature interfaces and supporting software. [22, 24]	Battery-box construction/CAD, Kapton/Teflon and orbital temperature datasets. [87, 106] BIRDS-3 EM/Uguisu FM and BIRDS-4 FM TVAC reports, measured profiles and battery-cell thermal screening documentation. [89, 90, 107] *BIRDS-3 orbital data used no heater. EM heating slowed cooling without holding 0 degrees C. Test records retain profile/channel inconsistencies.* FAB heater/temperature schematic, BOM and threshold-control code. [108] BIRDS-5 FAB firmware with automatic heater control commented out for power saving. [29] TVAC procedures and SAT1_FM raw temperature/pressure logs. [92] *Draft tables and channel mapping remain incomplete. Logs are laboratory data.*
	Heater/sensor schematics, ChibiOS heater control and thermal reorientation source. [4, 8, 99] *Reorientation retains the C3 wheel-command TODO.* Battery/Zephyr configuration and engineering notes. [100] *Default heaters are disabled.*					
Propulsion	No thruster/feed CAD, schematics, BOM or propulsion control code identified in scope. [2]	No propulsion CAD, thruster/feed schematic, BOM or control source identified in scope. [60]	No propulsion CAD, thruster/feed schematic, BOM or control source identified for FloripaSat-I. [13, 62]	No propulsion CAD, feed/thruster electronics, BOM or control source identified. [16]	No propulsion CAD, feed/thruster electronics, BOM or control source identified. The AOCS manual plans no maneuvers. [72]	No propulsion CAD, feed/thruster schematic, BOM or control code identified in the scoped platform release. [27]

Subsystem	OreSat	UPSat	FloripaSat-I	PyCubed	AcubeSAT	BIRDS
Power	Solar v5.3 and battery v3 EAGLE schematics/PCBs/XLSX BOMs, MPPT/protection/switching firmware and assembly references. [2, 99, 109] *Fuel-gauge learning cycles were skipped on OreSat0 and OreSat0.5. The December 2024 update reports slowly discharging OreSat0.5 batteries. [55, 100]*	Altium EPS v6 schematic/PCB and PDF, MPPT/regulation/switching/protection firmware, solar/battery models and intended flight releases. [59, 82, 101] *The EPS schematic names STM32L152. Its README names STM32L100RCT6.*	EPS schematic PDF, MPPT/energy/protection firmware and battery/purchased-panel descriptions. [13, 103, 104] *The EPS wiki specifies five GomSpace P110 panels and one ISIS top panel.*	Platform power-design publication, historical circuitry and flight battery/energy results. [14–16] *V-R3x harvested roughly one tenth of expected solar power and remained power negative.*	2019 KiCad EPS schematic/PCB and BOM.xlsx. [23, 72, 120] *The CDR architecture uses commercial EPS/battery modules. The 2019 prototype retains design TODOs.*	OBC/EPS/FAB schematics/BOM/firmware, native solar/backplane/access-board source and BIRDS-3/4 EPS datasets. [25, 106, 108, 121]
	Battery v4.1, MCXN947 solar KiCad designs, Zephyr applications and engineering notes. [100, 110]		Bench MPPT/heater experiments, Simulink/Simscape and LabVIEW test/model artifacts. [119]	Power.sch, protected battery-board schematic/PCB/BOM, charger/MPPT/monitoring drivers and manufacturing files. [17, 18, 83] *Flight setup calls for removing USB charging jumpers and selecting solar diode D2.*	Power-budget scripts/STK models and documented purchased EPS, battery and array architecture. [23, 72]	BIRDS-4 measured power budget, current calibration and EPS test setup/results presentation. [122] *The BIRDS-4 report records 458 days of Tsuru operation and power-related loss of Maya-2 and GuaraniSat-1 within one week. [26]*
	2021 solar MPPT algorithm, circuit and EPS integration design paper. [111] *Exact mapping to flight solar v5.3 is unverified.*			Beep-Sat battery task and low-power/shutdown example source. [20, 48] *The shutdown example leaves wakeup behavior as an exercise.*		Separate firmware, parts lists and test/procedure package. [29, 34, 123] *BIRDS-5 OBC/EPS and FAB BOMs retain BIRDS-3 revision names.*
	Battery-cell ground tester KiCad/BOM/wiring, prototype control/logging and design poster. [112, 113] Solar-simulator v2.1 fabrication/assembly files, Pico headless2 firmware/CLI and design/protocol documentation. [114–118] *Equipment revisions differ. Completed cell qualification and spectral calibration data are unverified.*					Solar-fault preprocessing/model documentation and Raspberry Pi 2 inference benchmarks. [124] 2026 reliability paper includes simulated mixed-cell BIRDS-3/4 arrays. [125] *No onboard detector deployment or modified BIRDS-3/4 flight arrays are established.*

Table 2: Ground segment materials. Contribution groups are colored by status: Flight-linked , Changed or superseded , Unvalidated , Ground reference , Not found .

Subsystem	OreSat	UPSat	FloripaSat-I	PyCubed	AcubeSAT	BIRDS
Planning and scheduling	UniClOGS pass predictor/commander, pass-calculator code, tests and the published station architecture. [126, 127] *Inter-station coordination is manual.* RADS and PAWS request/booking interfaces. [128, 129] *RADS forwarding is marked TODO. PAWS request lookup remains unfinished.*	SatNOGS observation-job fetching and pass-scheduling source plus architecture documentation. [60, 130] Archived JavaScript commands for onboard scheduling and science-script control. [130] *The UPSat web extension was removed upstream in May 2017.*	Time-window/periodicity uplink_event.cpp, timed dispatch and Gpredict launch integration. [131]	Context only: tracking/operations publications and onboard tasko examples. [16, 48]	Yamcs ST[11] command definitions, release-time form and documented SatNOGS pass scheduling. [132–134] *The monitoring repository contains one sample DAG with schedule=None.*	Context only: tracking functions and onboard reservation firmware. [25, 135]
Spacecraft commanding	Yamcs shell/XTCE/YAML, EDL postprocessor, pass commander transport and matching C3 command handlers. [126, 136, 137] 2022 poster documents FlatSat command/response verification tests. [138] *Raw test logs and the tested source revision were not recovered.*	ECSS command builder, browser UI, packet/large-data handlers, serial/UDP transport and historical transmit integration. [60, 130, 139] *The UPSat-specific SatNOGS client extension was removed upstream in May 2017.*	Qt NGHam command generation/dispatch, GNU Radio GFSK transmit flowgraph, Makefile and matching spacecraft handlers. [12, 13, 131] Published serial/server output code paths and GUI hooks. [131] *Selecting serial or server uplink raises an Under development error.*	Mission paper describing successful ground command/control and station architecture. [16] Ground-radio command/ACK examples and spacecraft cdh.py handlers. [20, 48] *Command examples use fixed passcodes and remote eval/exec.*	Yamcs postprocessor/XTCE/history, Angular validation/submission and serial/TCP bridges. [132, 133]	C# command UI, Transmit_async, ADDNICS/KISS/AX.25 framing and TNC control. [27, 135] Separate BIRDS-5 v1.5.0 operator manual: 11-byte commands, ACK retry and CW/FM procedures. [140, 141] *That application package is binary-only. The manual establishes no later spacecraft contact.*
Ground equipment control	Stationd GPIO/PA/LNA/relay control, rotator/Doppler/safing source and tests, UPD KiCad/wiring and relay-driver BOM. [127, 142, 143] UniClOGS construction manual, native station/rack CAD, editable wiring and commissioning procedures for power, rotator, relay, LNA and RF-path checks. [144] *The 2024 paper documents operating stations. The manual remains a partial draft. CAD exports and restricted linked BOM/SOP files remain unverified.*	Rotctld/rigctld observer workers, tracking/Doppler control and station setup documentation. [130]	SDR device/frequency/rate controls, GNU Radio configuration and Gqrx/Gpredict launch integration. [131]	V-R3x telescope-mount/antenna and tracking setup publication. [16] radio_helpers.py device configuration and receiver sleep/MQTT control code. [145] *Remote MQTT commands invoke eval/exec.*	Specialized SatNOGS rotator/EasyComm/homing/end-stop/motor firmware and integrated upstream hardware references. [72, 146]	Prosistel/Yaesu, radio mode/frequency/Doppler and TLE tracking source with textbook setup. [27, 135] BIRDS-2 UiTM fault-isolation, CI-V fallback and surge/shutdown procedures. [147] BIRDS-5 power-up/TLE checklist and supporting TLE retrieval script. [140, 141]

Subsystem	OreSat	UPSat	FloripaSat-I	PyCubed	AcubeSAT	BIRDS
Space/ground communications	GNU Radio TX/RX flowgraphs, SDR/station code, interface electronics and RF wiring/ configuration documentation. [127, 143, 148] *The RF chain uses commercial power amplifiers and SDR equipment.*	USRP TX/RX flowgraphs, AX.25/QB50 framing/deframing and UDP transport source/ configuration. [130, 139] *The transceiver script uses Python 2.*	GFSK TX/RX flowgraphs, WAV/ UDP reception, sample recordings, support list and reception instructions. [45, 131]	Published helical antenna, LNA, amplifier and bidirectional mission link description. [16]	CDR RF chain/LimeSDR/antenna design, uplink GMSK GNU Radio flowgraph and Yamcs frame interfaces. [72, 132, 151] *The UHF-downlink and S-band flowgraph directories contain README files only.*	KISS/AX.25/TNC serial software, Icom/KPC9612+/Yagi configuration and documented SatNOGS reception. [27, 34, 135]
	Handheld receiver mechanical/ electronic source, firmware and assembly material. [40]		GRS NGHam framing/Reed-Solomon source. [131] *Reed-Solomon initialization sets nroots, mm and nn to zero.*	PyGS shopping list, wiring, assembly, spacer STL and receive code, plus Beep-Sat transmit examples. [20, 145] *PyGS publishes third-party library licenses only.*		
	FlatHILS RF test poster and Basilisk access/range-to-USB attenuation source, driver and example scripts on new_sim. [149, 150] *The later code is not a verified poster build. Examples can run without hardware. End-to-end RF results and Doppler simulation remain unverified.*					
Mission data processing	Handheld FEC/block reconstruction C code, receive scripts, MP4 handling and ffprobe stream checks. [40, 152] *Receiver documentation reports double-free and invalid-pointer crashes, handled by service restarts.*	Historical ECSS raw science-unit extraction, large-data reassembly, filenames and UTC handling. [130] *The science-log parser belongs to the removed UPSat-specific client extension.*	telemetry_data.cpp payload bytes and experiment-specific payload_ x_status.cpp source. [131]	Dynamics/measurement formulation and Levenberg-Marquardt pseudocode in the V-R3x paper. [16] *Complete ranging and radiation datasets remained onboard after operations ended.*	Bench UI raw-camera image assembly, instrument plots and tests. [153] Yamcs TMParser.java segment reconstruction and custom PUS preprocessing. [132]	Context only: published mission images and raw-hex capture/ logging. [26, 135]
Mission data dissemination	Published telemetry web endpoint and mission ground data-access architecture. [127, 154]	SatNOGS recording/waterfall upload source and public observation/telemetry references. [60, 130]	Public audio/data/image archive, SatNOGS observation references and GRS packet/CSV export code. [131, 155]	Context only: publications, raw MQTT relay and beacon-decoder links. [16, 145]	Angular bucket/file/image dialogs, list/upload/download/delete and Yamcs object/archive APIs. [132, 133]	Context only: published images, technical papers and actual EPS datasets. [26, 106, 135]
	OreSat Live web/video server/ player and PAWS frontend. [128, 152]					
Network	Ethernet/VLAN/VPN architecture, station service/image setup, systemd network files and TCP/ UDP interfaces. [127, 137, 156] *The 2024 paper describes deployed station networking.*	HTTP job/upload API, TCP Hamlib and UDP radio source, with architecture/install documentation. [130]	udp_server.cpp and udp_ decoder.cpp source. [131]	WiFi/socket/MQTT publish-subscribe source, offline cache and later upload logic. [145] *The example uses unencrypted MQTT and an external broker.*	Yamcs TCP/HTTP/WebSocket connections, mission YAML, Docker instructions and serial/ TCP bridges. [132, 133]	BIRDS network/SatNOGS descriptions. The released C# application mainly uses local files/serial devices. [34, 135] Earlier BIRDS-1 thesis summary documents central-server architecture and preflight transfer/storage integration tests. [157] *Server source and executed distributed scheduling were not verified.*
		Redis/Socket.IO transport belonging to the historical UPSat web interface. [130] *The web-specific transport was removed from upstream SatNOGS.*	Published external-server uplink hooks. [131] *The external-server uplink path raises an Under development error.*			

Subsystem	OreSat	UPSat	FloripaSat-I	PyCubed	AcubeSAT	BIRDS
Telemetry processing	Beacon schemas/decoders, SatNOGS import, Yamcs preprocessing/tests, XTCE databases, archive and display configuration. [136, 137, 158]	ECSS/WOD decoding and engineering conversion in the historical UPSat client extension. [60, 130] *The UPSat decoder was removed from upstream SatNOGS.* Standalone WOD plotting source and example JSON data. [159] *Example datasets are ground-test data dated 2016.*	Packet/engineering conversion, display/logging, CSV plotting and sample recordings. [13, 131] GRS FEC implementation and standalone Python decoder helpers/GUI. [131, 160] *The standalone decode callback only loads a WAV file and logs its sample rate.*	Published V-R3x beacon layout and reception/telemetry results. [16, 105] CRC/hex/JSON handling with time/station/RSSI metadata. [145]	XTCE databases, preprocessing, recorder/alarm/replay configuration and live/archive graphs. [132, 133] New archiving-service template and monitoring example DAG. [134, 161]	Actual EPS datasets and calibration/processing publication. [106] BIRDS-4 CW bit specification, operator guide and editable conversion workbook with labelled reception records. [141, 162] *CW decoder download is binary-only. Workbook calibration was not independently validated.* BIRDS-RPM telemetry test dashboard source/manuals: serial input, log replay and CSV export. [163] *Released packet definitions are generic demos.* C# serial/KISS intake, hex conversion, packet counts and log export. [135]

Table 3: Sources cited in the materials tables. Repository URLs identify the reviewed reference or branch where available.

Ref.	Source
[1]	OreSat Project. oresat-c3-hardware @ v5_0, . URL https://github.com/oresat/oresat-c3-hardware/tree/v5_0. Accessed 3 November 2026
[2]	Portland State Aerospace Society. OreSat0: mission history and subsystem manifest, . URL https://www.oresat.org/satellites/oresat0. Accessed 3 November 2026
[3]	OreSat Project. oresat-c3-hardware @ 738a430, . URL https://github.com/oresat/oresat-c3-hardware/tree/738a430a2270f489b3ff0aad043578f60fcf075e. Accessed 3 November 2026
[4]	OreSat Project. oresat-c3-software @ 1136e31, . URL https://github.com/oresat/oresat-c3-software/tree/1136e319fb85844638c0c549663045f731468e7f. Accessed 3 November 2026
[5]	Portland State Aerospace Society. *OreSat Software, Release 1.0*, 2026. URL https://oresat-software.readthedocs.io/en/latest/. Living manual, PDF built 9 June 2026. Accessed 4 November 2026
[6]	OreSat Project. oresat-linux @ fc06a67, . URL https://github.com/oresat/oresat-linux/tree/fc06a67b7894cf1864470314e5f5da54c85a3c14. Accessed 4 November 2026
[7]	OreSat Project. oresat-olaf @ 7b32e7a, . URL https://github.com/oresat/oresat-olaf/tree/7b32e7a088edae80bce5a3d2a26122408c96c655. Accessed 4 November 2026
[8]	OreSat Project. oresat-firmware @ c3_v1_1, . URL https://github.com/oresat/oresat-firmware/tree/c3_v1_1/src/f4/app_control/source. Accessed 3 November 2026
[9]	Libre Space Foundation. upsat-obc-hardware @ 4661719, . URL https://gitlab.com/librespacefoundation/upsat/upsat-obc-hardware/-/tree/4661719549217c3c32389e0c80aeca8c0dc1eb12. Accessed 3 November 2026
[10]	Libre Space Foundation. upsat-obc-software @ a04902f, . URL https://gitlab.com/librespacefoundation/upsat/upsat-obc-software/-/tree/a04902fc894e9c22d39a4b5e92af9874f256e941. Accessed 3 November 2026
[11]	Libre Space Foundation. upsat-ecss-services @ 616003e, . URL https://gitlab.com/librespacefoundation/upsat/upsat-ecss-services/-/tree/616003e52e2ab282054b89c1dfac0b4bd0f96e1b. Accessed 3 November 2026
[12]	FloripaSat Team. obdh @ e24bf3e, . URL https://github.com/floripasat/obdh/tree/e24bf3e6395eafdae88358c90d23c096502167ff. Accessed 3 November 2026
[13]	Qualification and validation test methodology of the open-source CubeSat FloripaSat-I, 2020. URL https://www.jseepub.com/article/2020/1004-4132/20200615.shtml. Accessed 3 November 2026
[14]	PyCubed Project. hardware @ d1adfd0, . URL https://github.com/pycubed/hardware/tree/d1adfd0c207dd5951ab175af53b3b7c1c72a5212/mainboard-v01. Accessed 3 November 2026
[15]	PyCubed Project. PyCubed: An Open-Source, Radiation-Tested CubeSat Platform Programmable Entirely in Python, 2019. URL https://github.com/pycubed/documentation/blob/0670db47f1a26204e200c8a92beb2f431a1d65cb/PyCubed_smallsat-paper.pdf. Accessed 3 November 2026
[16]	The V-R3x Mission: Towards Autonomous Networking and Navigation for CubeSat Swarms, 2022. URL https://roboticexplorationlab.org/papers/vr3x.pdf. Accessed 3 November 2026
[17]	PyCubed Project. hardware @ d1adfd0, . URL https://github.com/pycubed/hardware/tree/d1adfd0c207dd5951ab175af53b3b7c1c72a5212/mainboard-v05. Accessed 3 November 2026
[18]	PyCubed Project. software @ 72ab3a1, . URL https://github.com/pycubed/software/tree/72ab3a1e1869b4d8a6096102e156567f89b0059b/mainboard-v05. Accessed 3 November 2026
[19]	PyCubed Project. NOTICE: August 1 2024, SAMD51 proton sensitivity, 2024. URL https://github.com/pycubed/forums/discussions/87. Accessed 3 November 2026
[20]	PyCubed Project. software_example_beepsat @ c057fa0, . URL https://github.com/pycubed/software_example_beepsat/tree/c057fa050b28df7fc0a898416409a4c1dc8d325d. Accessed 3 November 2026
[21]	AcubeSAT Team. obc-pcb @ adcs-obc-v2.5, . URL https://gitlab.com/acubesat/obc/obc-pcb/-/tree/adcs-obc-v2.5. Accessed 3 November 2026
[22]	AcubeSAT Team. embedded-software @ main, . URL https://gitlab.com/acubesat/obc/embedded-software/-/tree/main. Accessed 3 November 2026
[23]	Lessons learned from the design and early manufacturing phases of AcubeSAT, 2026. URL https://mediatum.ub.tum.de/doc/1857487/1857487.pdf. Accessed 3 November 2026
[24]	AcubeSAT Team. obc-pcb @ v3.1, . URL https://gitlab.com/acubesat/obc/obc-pcb/-/tree/v3.1. Accessed 3 November 2026
[25]	BIRDS Open Source Project. BIRDS4-OBC @ 490f744, . URL https://github.com/BIRDSOpenSource/BIRDS4-OBC/tree/490f74487d65ebfb0037bbdd9f3e7aab6aefc3a5. Accessed 3 November 2026
[26]	BIRDS-4 1U CubeSat Missions and On-Orbit Results. URL https://nanosat11th.itu.edu.tr/assets/papers/BIRDS-4.pdf. Accessed 3 November 2026
[27]	Kyushu Institute of Technology. BIRDS Platform Interface Control Document, English Rev F. URL https://github.com/BIRDSOpenSource/BIRDS-GeneralDocumentation/blob/main/BIRDS_Platform_ICD_ENGLISH_RevF_0303.pdf. Accessed 3 November 2026
[28]	Marloun Sejera. Configurable Backplane, 2022. URL https://birds-project.com/open-source/pdf/20220713_BIRDS_OpenSource_Backplane.pdf. BIRDS webinar 8, 13 July 2022. Accessed 4 November 2026
[29]	BIRDS Open Source Project. BIRDS5-Source_Codes @ 8dc0580, . URL https://github.com/BIRDSOpenSource/BIRDS5-Source_Codes/tree/8dc0580ba154e26f59eb1dfcd5d3bcc1e233c8ee. Accessed 3 November 2026
[30]	BIRDS-5 Project. *BIRDS-5 OBC Software Manual*, 2022. URL https://github.com/BIRDSOpenSource/BIRDS5-Source_Codes/blob/8dc0580ba154e26f59eb1dfcd5d3bcc1e233c8ee/Software%20Manual/BIRDS-5%20OBC%20Manual.docx. Initial release 7 July 2022, revision log credits Keenan. Accessed 4 November 2026
[31]	Keenan Chatar. BIRDS-5 OBC, 2022. URL https://birds-project.com/open-source/pdf/20220810_BIRDS_Open_Source_Webinar-OBC.pdf. BIRDS webinar 9, 10 August 2022. Accessed 4 November 2026
[32]	BIRDS Open Source Project. BIRDSX-OBC-FAB @ 57accd8, . URL https://github.com/BIRDSOpenSource/BIRDSX-OBC-FAB/tree/57accd8249b357ee59315ff869c4bd8d56184141. PIC host integration source. Accessed 4 November 2026
[33]	BIRDS Open Source Project. BIRDSX-PCB_Design @ f3ce3fd, . URL https://github.com/BIRDSOpenSource/BIRDSX-PCB_Design/tree/f3ce3fd57253c5ef947a0cbde3b548122e5a457f. Schematic PDFs and BOMs. Accessed 4 November 2026

Ref.	Source
[34]	BIRDS Project. Lessons Learnt and Findings from BIRDS5-FTA, 2023. URL https://birds-project.com/open-source/pdf/20230308_Birds5_project_OpenSourceWebinar.pdf. Accessed 3 November 2026
[35]	OreSat Project. oresat-at86rf215-driver @ 692bf07, 2026. URL https://github.com/oresat/oresat-at86rf215-driver/tree/692bf077d07bca4c29723d7a41dad0cce8387f3c. Accessed 4 November 2026
[36]	Hessah Alsaadon, Anna Meneely, Bryan Borgeson, and Alex Miller. Trailblazing Next Generation Digital Radios for Amateur Radio Cubesats, 2026. URL https://spacegrant.oregonstate.edu/sites/spacegrant.oregonstate.edu/files/2026-06/halsaadon_utep_posterpdf_spring-2026.pdf. Oregon NASA Space Grant Spring Symposium poster. Accessed 4 November 2026
[37]	OreSat Project. oresat-dxwifi-hardware @ dbe3558, . URL https://github.com/oresat/oresat-dxwifi-hardware/tree/dbe3558af5239e278f92e747b6ec1b2ab5582529. Spacecraft Live card, BOM and link models. Accessed 4 November 2026
[38]	OreSat Project. oresat-dxwifi-software @ ba411a0, . URL https://github.com/oresat/oresat-dxwifi-software/tree/ba411a0a3498421e84b215fc85a26af9d3c591b8. Accessed 4 November 2026
[39]	OreSat Project. oresat-libdxwifi @ c789aa1, . URL https://github.com/oresat/oresat-libdxwifi/tree/c789aa19e7035f4ecd24a1224784f84fcdd9668e. Accessed 4 November 2026
[40]	OreSat Project. oresat-live-handheld-ground-station, . URL https://github.com/oresat/oresat-live-handheld-ground-station. Accessed 3 November 2026
[41]	Libre Space Foundation. upsat-comms-hardware @ v3, . URL https://gitlab.com/librespacefoundation/upsat/upsat-comms-hardware/-/tree/v3. Accessed 3 November 2026
[42]	Libre Space Foundation. upsat-comms-hardware @ v3.1, . URL https://gitlab.com/librespacefoundation/upsat/upsat-comms-hardware/-/tree/v3.1. Accessed 3 November 2026
[43]	Libre Space Foundation. upsat-comms-software @ 7a4fcbd, . URL https://gitlab.com/librespacefoundation/upsat/upsat-comms-software/-/tree/7a4fcbdd6ed8eb2b34db2212b985304093fc17c9. Accessed 3 November 2026
[44]	Libre Space Foundation. upsat-comms-hardware @ a01575e, . URL https://gitlab.com/librespacefoundation/upsat/upsat-comms-hardware/-/tree/a01575e8586e57e83f77f50ba8cac69306b8650c. Accessed 3 November 2026
[45]	FloripaSat Team. ttc @ e697c85, . URL https://github.com/floripasat/ttc/tree/e697c85eb27b3a5de8aba4432b7f32a1ec5931cb. Accessed 3 November 2026
[46]	Daniel Estévez. Decoding FloripaSat-1, 2019. URL https://destevez.net/2019/12/decoding-floripasat-1/. Accessed 3 November 2026
[47]	SpaceLab UFSC. ttc @ master. URL https://github.com/spacelab-ufsc/ttc/tree/master/hardware. Accessed 3 November 2026
[48]	PyCubed Project. pycubed.github.io @ fd6951b, . URL https://github.com/pycubed/pycubed.github.io/tree/fd6951bd1a6d51d5c1edd02c7ab69ccb8eaca214. Accessed 3 November 2026
[49]	AcubeSAT Team. satnogs-comms-hardware @ frontend-routing, . URL https://gitlab.com/acubesat/comms/hardware/satnogs-comms-hardware/-/tree/frontend-routing. Accessed 3 November 2026
[50]	AcubeSAT Team. comms-sw @ main, . URL https://gitlab.com/acubesat/comms/software/comms-sw/-/tree/main. Accessed 3 November 2026
[51]	BIRDS Open Source Project. BIRDS3-COM, . URL https://github.com/BIRDSOpenSource/BIRDS3-COM. Accessed 3 November 2026
[52]	Tharindu Dayarathna. Overview of the BIRDS Satellite Communication Subsystem, Design of a Low-Cost UHF Transceiver, and Proposal for a Simplified OBC Architecture, 2025. URL https://birds-project.com/open-source/pdf/birds_bus_opensource_webinar_28.pdf. BIRDS webinar 28, 6 August 2025. Accessed 4 November 2026
[53]	BIRDS Open Source Project. BIRDSX-COM @ e2853ce, . URL https://github.com/BIRDSOpenSource/BIRDSX-COM/tree/e2853ce550576c4aecc9d2175549dad2c097db05. Antenna chamber measurements and tuning plots. Accessed 4 November 2026
[54]	OreSat Project. oresat-adcs-hardware @ 07febe9, . URL https://github.com/oresat/oresat-adcs-hardware/tree/07febe977d3965adc3d9214d350480048879b9fa. Accessed 4 November 2026
[55]	Portland State Aerospace Society. OreSat0.5: mission history, . URL https://www.oresat.org/satellites/oresat0-5. Accessed 3 November 2026
[56]	OreSat Project. oresat-reaction-wheel-app @ cf16c97, 2026. URL https://github.com/oresat/oresat-reaction-wheel-app/tree/cf16c975b78908018af7c501a81382da11c798d8. OreSat-Reaction-Wheel-ThesisWork branch. Accessed 4 November 2026
[57]	Chris Kane-Pardy. Advancing the OreSat ADCS: Design & Implementation of an Open-Source Reaction Wheel System. Master's thesis, Portland State University, August 2026. URL https://pdxscholar.library.pdx.edu/open_access_etds/7192/. Published 18 August 2026. Accessed 4 November 2026
[58]	Libre Space Foundation. upsat-adcs-hardware @ c9fddb4, . URL https://gitlab.com/librespacefoundation/upsat/upsat-adcs-hardware/-/tree/c9fddb419430afe172cc1a40d1c2f5837b582570. Accessed 3 November 2026
[59]	Libre Space Foundation. upsat-structural @ 4f5b03d, . URL https://gitlab.com/librespacefoundation/upsat/upsat-structural/-/tree/4f5b03d8727e7926ffb8648a3654af1a60df1907. Accessed 3 November 2026
[60]	UPSat technical thesis: on-board and command/control implementation, 2017. URL https://nchronas.github.io/upsat_msc_thesis/. Accessed 3 November 2026
[61]	Libre Space Foundation. upsat-adcs-software @ f3ff373, . URL https://gitlab.com/librespacefoundation/upsat/upsat-adcs-software/-/tree/f3ff373d05276bf6cd08a12626367f67d854a91b. Accessed 3 November 2026
[62]	FloripaSat Team. documentation, . URL https://github.com/floripasat/documentation/wiki/Satellite-General-Description. Accessed 3 November 2026
[63]	AcubeSAT Team. adcs-on-board-software, . URL https://github.com/AcubeSAT/adcs-on-board-software. Accessed 3 November 2026
[64]	Dulani Chamika Withanage, Mariko Teramoto, and Mengu Cho. On-Orbit Magnetometer Data Calibration Using Genetic Algorithm and Interchangeability of the Calibration Parameters. *Applied Sciences*, 13(11):6742, 2023. doi: 10.3390/app13116742. URL https://www.mdpi.com/2076-3417/13/11/6742. Accessed 4 November 2026
[65]	Dulani Withanage. On-orbit magnetometer data observed by BIRDS-3 CubeSats, 2023. URL https://data.mendeley.com/datasets/2b2rvxccyx/1. Version 1, 26 April 2023. Three XLSX workbooks. Accessed 4 November 2026
[66]	BIRDS-3 Project. BIRDS-3 ADCS Mission, 2020. URL https://birds1.birds-project.com/files/BIRDS_Newsletter_Issue_No.52.pdf. Newsletter 52, pp. 124–128, report by Dulani dated 15 May 2020. Accessed 4 November 2026

Ref.	Source
[67]	Timothy Kudzanayi Kuhamba. ADCS FM Test Up and FM Magnetometer Calibration, 2022. URL https://birds1.birds-project.com/files/BIRDS_Newsletter_Issue_No.72.pdf. Newsletter 72, pp. 110–125, presentations dated 20 January 2022. Accessed 4 November 2026
[68]	Ernesto Cortés. Introduction to BIRDS-RPM Project, 2026. URL https://birds-project.com/open-source/pdf/birds_bus_opensource_webinar_31.pdf. BIRDS webinar 31, 8 April 2026. Planned 3U generation. Accessed 4 November 2026
[69]	OreSat Project. oresat-gps-hardware @ 83b3ee2, . URL https://github.com/oresat/oresat-gps-hardware/tree/83b3ee21bde482f4e08fe5f95b72553dfa3021bf. Accessed 4 November 2026
[70]	OreSat Project. oresat-gps-software, . URL https://github.com/oresat/oresat-gps-software. Accessed 3 November 2026
[71]	OreSat Project. OreSat GPS Card: Fabrication and Assembly Information. Technical report, Portland State Aerospace Society, 2026. URL https://github.com/oresat/oresat-gps-hardware/blob/83b3ee21bde482f4e08fe5f95b72553dfa3021bf/oresat-gps-card/releases/oresat-gps-card-v1.2r1-2026-07-09T17-59-24.zip. Report and manufacturing files in v1.2r1 release ZIP, built 9 July 2026. Accessed 4 November 2026
[72]	AcubeSAT Team. cdr-public @ master, . URL https://gitlab.com/acubesat/documentation/cdr-public/-/tree/master. Accessed 3 November 2026
[73]	BIRDS-3 Project. BIRDS-3: Global Positioning System, 2020. URL https://birds1.birds-project.com/files/BIRDS_Newsletter_Issue_No.57.pdf. Newsletter 57, pp. 64–68, presentation by Dulani Chamika. Released 26 October 2020. Accessed 4 November 2026
[74]	Joven Javier. Subsystem Report-07: COTS GPS Mission, 2017. URL https://birds1.birds-project.com/files/BIRDS_Newsletter_Issue_No_15.pdf. Newsletter 15, pp. 56–63, 30 April 2017. Accessed 4 November 2026
[75]	Joven C. Javier. The BIRDS-2 GPS mission in brief, 2017. URL https://birds1.birds-project.com/files/BIRDS_Newsletter_Issue_No_21.pdf. Newsletter 21, pp. 50–53, 25 October 2017. Accessed 4 November 2026
[76]	OreSat Project. oresat-structure, . URL https://github.com/oresat/oresat-structure. Accessed 3 November 2026
[77]	Marvin Lin and Hayden Reinhold. OreSat Mechanical Architecture, 2021. URL https://events.libre.space/event/5/contributions/169/attachments/108/135/3-2%20OreSat%20Mechanical%20Architchure.pdf. Open Source CubeSat Workshop, 9 December 2021. Slides titled OreSat Mechanical Design. Accessed 4 November 2026
[78]	Kayla Rasavanh, Zachary Golden, Diana Hendricks, and Cass Blum. OreSat Structure Optimization, 2022. URL https://spacegrant.oregonstate.edu/sites/spacegrant.oregonstate.edu/files/zgolden_psu_uteap_poster_2022fallsymposium.jpg. Oregon NASA Space Grant Fall Symposium poster, original image. Linked PDF returns 404. Accessed 4 November 2026
[79]	Hayden Reinhold. OreSat Structures Intern, 2022. URL https://spacegrant.oregonstate.edu/sites/spacegrant.oregonstate.edu/files/2022-11/hreinhold_psu_orion_poster.pdf. Oregon NASA Space Grant Fall Symposium poster. Accessed 4 November 2026
[80]	OreSat Project. OreSat0.5: public Onshape assembly, . URL https://cad.onshape.com/documents/f25f3f200abb912b90d5ed0a/w/ea76d486620fd1b4122d88cc/e/4f39ca87568525b2426c52ab. Linked from the official CubeSat subsystems page. Document titled OreSat0.5 - DO NOT UPDATE. Accessed 4 November 2026
[81]	Cass Blum, Haohan Jiang, Danaya Murphy, and Teresa Nguyen. OreSat0.5: OSGC OrION Internships supporting Oregon's Second Satellite!, 2023. URL https://spacegrant.oregonstate.edu/sites/spacegrant.oregonstate.edu/files/2023-11/cblum_hjiang_tnyugen_dmurphy_psu_orion_poster.pdf. Oregon NASA Space Grant Fall Symposium poster. Accessed 4 November 2026
[82]	Libre Space Foundation. upsat-eps-software @ c03af5c, . URL https://gitlab.com/librespacefoundation/upsat/upsat-eps-software/-/tree/c03af5c43d8936f601537e5de85aaa5d2afee74f. Accessed 3 November 2026
[83]	PyCubed Project. hardware @ d1adfd0, . URL https://github.com/pycubed/hardware/tree/d1adfd0c207dd5951ab175af53b3b7c1c72a5212/batteryboard-v01b. Accessed 3 November 2026
[84]	AcubeSAT Team. antenna-deployment-mch @ master, . URL https://gitlab.com/acubesat/mechanisms/antenna-deployment-mch/-/tree/master. Accessed 3 November 2026
[85]	AcubeSAT Team. esatan-model @ main, . URL https://gitlab.com/acubesat/thermal/esatan-model/-/tree/main. Accessed 3 November 2026
[86]	AcubeSAT Team. AcubeSAT public assembly repository README, . URL https://gitlab.com/acubesat/structural/3d-cads/acubesat-3d-cads/public-3d-cads/acubesat-assembly-public/-/blob/main/README.md. Accessed 3 November 2026
[87]	BIRDS Open Source Project. BIRDS3-CAD, . URL https://github.com/BIRDSOpenSource/BIRDS3-CAD. Accessed 3 November 2026
[88]	BIRDS Open Source Project. BIRDS4-CAD, . URL https://github.com/BIRDSOpenSource/BIRDS4-CAD. Accessed 3 November 2026
[89]	BIRDS Open Source Project. BIRDS3-ProceduresAndReports @ c9ca7be, . URL https://github.com/BIRDSOpenSource/BIRDS3-ProceduresAndReports/tree/c9ca7be3c0cdaee0690f3a20af297f2dabce5413. Includes EM/FM vibration, antenna-release and thermal test reports. Accessed 4 November 2026
[90]	BIRDS Open Source Project. BIRDS4-ProceduresAndReports @ d7eadd9, . URL https://github.com/BIRDSOpenSource/BIRDS4-ProceduresAndReports/tree/d7eadd98f5327e1ddea82b640293d49bfdb09555. Includes assembly/rework manual, EM vibration report, FM test plans and thermal report. Accessed 4 November 2026
[91]	BIRDS Open Source Project. BIRDS5-CAD, . URL https://github.com/BIRDSOpenSource/BIRDS5-CAD. Accessed 3 November 2026
[92]	BIRDS Open Source Project. BIRDS5-Procedures_Reports @ e3f436f, . URL https://github.com/BIRDSOpenSource/BIRDS5-Procedures_Reports/tree/e3f436f8df9d3eb2ef43f0062c9034266421cd95. Accessed 4 November 2026
[93]	BIRDS Open Source Project. BIRDSX-CAD @ 36bceea, . URL https://github.com/BIRDSOpenSource/BIRDSX-CAD/tree/36bceeacc9f899477f2f10e36f8a65f7b58cb4ed. Separate 2U STEP, manufacturing drawings and calculation package. Accessed 4 November 2026
[94]	OreSat Project. oresat-thermal, . URL https://github.com/oresat/oresat-thermal. Accessed 3 November 2026
[95]	Parker Southwick, Jeremy Lowman, Katherine Popchock, Thomas Otero, Tyler Benson, and Griffin Johnson. OreSat Thermal Analysis: Progress Report, Winter 2019. Technical report, Portland State University, 2019. URL https://github.com/oresat/oresat-thermal/blob/568eba24c7932584c5dbf3659169a8f9bf1fd16b/Reports/ME492/ME492%20Progress%20Report.docx. Mechanical engineering capstone report. Accessed 4 November 2026

Ref.	Source
[96]	Tom Otero, Jeremy Lowman, Griffin Johnson, Tyler Benson, and Parker Southwick. Temperature Mapping of Conductive Heat Transfer Through a Printed Circuit Board. Technical report, Portland State University, 2019. URL https://github.com/oresat/oresat-thermal/blob/568eba24c7932584c5dbf3659169a8f9bf1fd16b/411_Measurements_Experiment/OreSat%20Measurements%20Report_.docx. ME 411 report, 18 March 2019. Accessed 4 November 2026
[97]	OreSat Project. oresat-thermal @ 568eba2, 2019. URL https://github.com/oresat/oresat-thermal/tree/568eba24c7932584c5dbf3659169a8f9bf1fd16b. gh-pages branch, historical MATLAB/ANSYS models and PCB conduction data. Accessed 4 November 2026
[98]	Michael Corcoran, Zachary Dilday, Oleg Krishcko, Danny Nelson, Tara Prevo, Catie Spivey, and Ian Robinson. Thermal Analysis and Testing of OreSat, 2020. URL https://spacegrant.oregonstate.edu/sites/spacegrant.oregonstate.edu/files/d7-thermal-analysis-and-testing-of-oregons-first-satellite.pdf. Oregon NASA Space Grant Symposium poster, 13 November 2020. Accessed 4 November 2026
[99]	OreSat Project. oresat-batteries @ v3.0, . URL https://github.com/oresat/oresat-batteries/tree/v3.0. Accessed 3 November 2026
[100]	OreSat Project. oresat-battery-app @ main, . URL https://github.com/oresat/oresat-battery-app/blob/main/README.md. Accessed 3 November 2026
[101]	Libre Space Foundation. upsat-eps-hardware @ 3dbe37e, . URL https://gitlab.com/librespacefoundation/upsat/upsat-eps-hardware/-/tree/3dbe37e58d44678d21044eccc7b0bb6fff322fd6. Accessed 3 November 2026
[102]	Libre Space Foundation. UPSat HeaterBoard design files, . URL https://gitlab.com/librespacefoundation/upsat/HeaterBoard/-/tree/20eece3b171fbb15dea49917cd1591856a9e3f32. Accessed 3 November 2026
[103]	FloripaSat Team. eps, . URL https://github.com/floripasat/eps/wiki/External-Hardware. Accessed 3 November 2026
[104]	FloripaSat Team. eps @ 3b4c252, . URL https://github.com/floripasat/eps/tree/3b4c252434d6a415e26725f7b8c308fa5d30a488. Accessed 3 November 2026
[105]	NASA. V-R3x S3VI Community of Practice webinar, 2021. URL https://www.nasa.gov/wp-content/uploads/2021/11/2021-11-10_v-r3x_s3vi_nov_cop.pdf. Accessed 3 November 2026
[106]	BIRDS Project. On-orbit electrical power system dataset of 1U CubeSat constellation, 2022. URL https://github.com/BIRDSOpenSource/EPS_dataset/tree/67ad4435c5329abde65afc16e30d725cd5f2653c. BIRDS-3 and Tsuru/BIRDS-4 datasets. Accessed 4 November 2026
[107]	BIRDS-4 Project. Battery (Cells) Screening Procedures and its Verification Report. Technical report, Kyushu Institute of Technology, 2020. URL https://github.com/BIRDSOpenSource/BIRDS3-ProceduresAndReports/blob/c9ca7be3c0cdaee0690f3a20af297f2dabce5413/Battery%28Cells%29%20Screening%20Procedures%20and%20its%20verification%20Report_V1.0%20%282%29.pdf. Revision 1, 4 September 2020, with a 2021 procedure addition. Report identifies BIRDS-4 despite repository location. Accessed 4 November 2026
[108]	BIRDS Open Source Project. BIRDS4-FAB, . URL https://github.com/BIRDSOpenSource/BIRDS4-FAB. Accessed 3 November 2026
[109]	OreSat Project. oresat-solar-hardware @ v5.3, . URL https://github.com/oresat/oresat-solar-hardware/tree/v5.3. Accessed 3 November 2026
[110]	OreSat Project. oresat-solar-hardware, . URL https://github.com/oresat/oresat-solar-hardware. Accessed 3 November 2026
[111]	David A. Lay, Grace Semerjian, Miles Simpson, Malay Das, and Andrew Greenberg. An Open Source and Modular CubeSat Solar Array with an Integrated Variable Step Incremental Conductance Maximum Power Point Tracker. In *AIAA Scitech 2021 Forum*, 2021. doi: 10.2514/6.2021-0804. URL https://arc.aiaa.org/doi/10.2514/6.2021-0804. AIAA 2021-0804. Accessed 4 November 2026
[112]	OreSat Project. oresat-battery-testing @ 6cc43c1, . URL https://github.com/oresat/oresat-battery-testing/tree/6cc43c1f3dffc2712cee1b59491adf140a256486. Accessed 4 November 2026
[113]	Benjamin Crall. OreSat Electrical, 2022. URL https://spacegrant.oregonstate.edu/sites/spacegrant.oregonstate.edu/files/2022-11/bcrall_psu_intern_poster.pdf. Oregon NASA Space Grant Fall Symposium poster, 10 November 2022. Accessed 4 November 2026
[114]	OreSat Project. oresat-solar-simulator-hardware @ a71b88f, . URL https://github.com/oresat/oresat-solar-simulator-hardware/tree/a71b88f613243dfbcb8cef192e0b9b718e1a0136/boards/controller-board/releases. Includes controller v2.1 fabrication/assembly release. Accessed 4 November 2026
[115]	OreSat Project. oresat-solar-simulator-software @ 521c99c, 2026. URL https://github.com/oresat/oresat-solar-simulator-software/tree/521c99c25cb7888ef7e2ef78753614a67d336f6b. headless2 branch. Accessed 4 November 2026
[116]	Adam Martinez and Cesar Ordaz-Coronel. OreSat Solar Simulator, 2023. URL https://spacegrant.oregonstate.edu/sites/spacegrant.oregonstate.edu/files/2023-11/amartinez_psu_uteap_poster.pdf. Oregon NASA Space Grant Fall Symposium poster, 9 November 2023. Accessed 4 November 2026
[117]	Daniel Monahan, Savannah Bedford, and James McNames. OreSat Testing Infrastructure: Making NASA CubeSat Launch Initiative (CSLI) Programs More Reliable, 2024. URL https://spacegrant.oregonstate.edu/sites/spacegrant.oregonstate.edu/files/2024-10/sbedforddmonahan_psu_uteap_poster.pdf. Oregon NASA Space Grant Fall Symposium poster, 11 October 2024. Accessed 4 November 2026
[118]	Bryan Borgeson. FlatHILS Hardware-in-the-Loop Testing and Orchestration Framework, 2026. URL https://spacegrant.oregonstate.edu/sites/spacegrant.oregonstate.edu/files/2026-10/bborgeson_psu_orion_poster_fall-2026_0.pdf. Oregon NASA Space Grant Fall Symposium poster, 9 October 2026. Accessed 4 November 2026
[119]	FloripaSat Team. eps-tests, . URL https://github.com/floripasat/eps-tests. Accessed 3 November 2026
[120]	AcubeSAT Team. eps-pcb @ master, . URL https://gitlab.com/acubesat/eps/eps-pcb/-/tree/master. Accessed 3 November 2026
[121]	BIRDS Open Source Project. BIRDS4-SolarPanel, . URL https://github.com/BIRDSOpenSource/BIRDS4-SolarPanel. Accessed 3 November 2026
[122]	Hari Ram Shrestha. Power Budget for 1U Satellite, 2022. URL https://birds-project.com/open-source/pdf/Power-Budget-Analysis-for-1U-satellit20220514.pdf. BIRDS webinar 6, 14 May 2022. Accessed 4 November 2026
[123]	BIRDS Open Source Project. BIRDS5-Parts_list, . URL https://github.com/BIRDSOpenSource/BIRDS5-Parts_list. Accessed 3 November 2026

Ref.	Source
[124]	Adolfo Javier Jara Cespedes, Bramandika Holy Bagas Pangestu, Akitoshi Hanazawa, and Mengu Cho. Performance Evaluation of Machine Learning Methods for Anomaly Detection in CubeSat Solar Panels. *Applied Sciences*, 12(17):8634, 2022. doi: 10.3390/app12178634. URL https://www.mdpi.com/2076-3417/12/17/8634. Accessed 4 November 2026
[125]	Hari Ram Shrestha, Jose Rodrigo Cordova-Alarcon, Necmi Cihan Örger, Victor Hugo Schulz, Fahd Moumni, Takashi Yamauchi, and Mengu Cho. Improving the Reliability of a CubeSat Power System by Combining Small-Sized Monocrystalline Cells With Triple-Junction Solar Cells. *IEEE Access*, 14:25038–25054, 2026. doi: 10.1109/ACCESS.2026.3664750. URL https://ieeexplore.ieee.org/document/11395955/. Modified BIRDS-3/4 arrays are simulations. Flight results concern MicroOrbiter-1/CURTIS. Accessed 4 November 2026
[126]	UniClOGS Project. uniclogs-pass-commander, . URL https://github.com/uniclogs/uniclogs-pass-commander. Accessed 3 November 2026
[127]	University Class Open Ground Station (UniClOGS), 2024. URL https://doi.org/10.26077/8jm6-9n70. Accessed 3 November 2026
[128]	OreSat Project. oresat-paws, . URL https://github.com/oresat/oresat-paws. Accessed 3 November 2026
[129]	UniClOGS Project. rads @ a454091, . URL https://github.com/uniclogs/rads/tree/a4540910af859f025c33a777cf6ef5708181c96b. Accessed 3 November 2026
[130]	Libre Space Foundation. satnogs-client @ 264dcb0, . URL https://gitlab.com/librespacefoundation/satnogs/satnogs-client/-/tree/264dcb04b3ea4c34f2b05d7b7e89df83cf29a328. Accessed 3 November 2026
[131]	FloripaSat Team. grs @ c6c9659, . URL https://github.com/floripasat/grs/tree/c6c9659fca7c655c94e286cca9f7860c139e8889. Accessed 3 November 2026
[132]	AcubeSAT Team. yamcs-instance @ main, . URL https://gitlab.com/acubesat/ops/yamcs-instance/-/tree/main. Accessed 3 November 2026
[133]	AcubeSAT Team. frontend-software @ main, . URL https://gitlab.com/acubesat/ops/frontend-software/-/tree/main. Accessed 3 November 2026
[134]	AcubeSAT Team. monitoring-service @ main, . URL https://gitlab.com/acubesat/ops/monitoring-service/-/tree/main. Accessed 3 November 2026
[135]	BIRDS Open Source Project. BIRDS-GS_software @ 780c81c, . URL https://github.com/BIRDSOpenSource/BIRDS-GS_software/tree/780c81cf86be74bb3cd238b394586425f2cddfa9. Accessed 4 November 2026
[136]	UniClOGS Project. yamcs, . URL https://github.com/uniclogs/yamcs. Accessed 3 November 2026
[137]	UniClOGS Project. uniclogs-configs, . URL https://github.com/uniclogs/uniclogs-configs. Accessed 3 November 2026
[138]	Kyle Esquerra. UniClOGS: Ground Station Commanding, 2022. URL https://spacegrant.oregonstate.edu/sites/spacegrant.oregonstate.edu/files/2022-11/kesquerra_psu_orion_intern_poster.pdf. Oregon NASA Space Grant Fall Symposium poster, 10 November 2022. Accessed 4 November 2026
[139]	Libre Space Foundation. gr-satnogs @ ca1d744, . URL https://gitlab.com/librespacefoundation/satnogs/gr-satnogs/-/tree/ca1d744d435a482a3b87f1c6a803967be1fbc2c7. Accessed 3 November 2026
[140]	Edgar Mujuni and Ramson Nyamukondiwa. *BIRDS GS Operation Manual, Version 1*, 2022. URL https://github.com/BIRDSOpenSource/BIRDSX-GS/blob/c5ae7855d0f0576d70ec2fe3c92f1d1cde4c16d8/BIRDS%20GS%20Software/Reference/BIRDS5%20GS%20Operation%20Manual_V1.pdf. 1 December 2022. BIRDS-5 operator application v1.5.0. Accessed 4 November 2026
[141]	BIRDS Open Source Project. BIRDSX-GS @ c5ae785, . URL https://github.com/BIRDSOpenSource/BIRDSX-GS/tree/c5ae7855d0f0576d70ec2fe3c92f1d1cde4c16d8. Includes BIRDS-4 CW workbook, BIRDS-5 manual, binaries and TLE retrieval script. Accessed 4 November 2026
[142]	UniClOGS Project. uniclogs-stationd, . URL https://github.com/uniclogs/uniclogs-stationd. Accessed 3 November 2026
[143]	UniClOGS Project. uniclogs-hardware, . URL https://github.com/uniclogs/uniclogs-hardware. Accessed 3 November 2026
[144]	Kaleb Rice, Anna Meneely, Glenn LeBrasseur, and Andrew Greenberg. *Everything You Need To Know But Were Too Afraid To Ask To Build Your Own Ground Station: UniClOGS and You*. URL https://docs.google.com/document/d/1X3NJvZIJBoTSr9gLLe9Uswm0GSTiX_2NS_AyfrqKy0U/edit. Live working draft. Title page updated 28 November 2023, with later 2024 annotations. Accessed 4 November 2026
[145]	maholli. PyGS @ 34f95b8. URL https://github.com/maholli/PyGS/tree/34f95b88323805511a2acc71627d0d8ddc609c78. Accessed 3 November 2026
[146]	AcubeSAT Team. antenna-rotator-software @ main, . URL https://gitlab.com/acubesat/comms/ground-station/antenna-rotator-software/-/tree/main. Accessed 3 November 2026
[147]	F. Z. Ali, S. N. M. Rahim, and M. H. Jusoh. Amateur satellite ground station: Troubleshooting and lesson learned. *Journal of Physics: Conference Series*, 1768:012013, 2021. doi: 10.1088/1742-6596/1768/1/012013. URL https://iopscience.iop.org/article/10.1088/1742-6596/1768/1/012013. UiTMSAT-1/BIRDS-2 ground station. Accessed 4 November 2026
[148]	UniClOGS Project. uniclogs-sdr @ maint-3.10, . URL https://github.com/uniclogs/uniclogs-sdr/tree/maint-3.10. Accessed 3 November 2026
[149]	Max Eltzroth, Hannah Rose Edington, Nicole Blase, and Savannah Krupa. FlatHILS: FlatSat Hardware-in-the Loop Simulation Infrastructure for Small Satellite Labs, 2025. URL https://spacegrant.oregonstate.edu/sites/spacegrant.oregonstate.edu/files/2025-08/meltzroth_psu_afrip_poster.pdf. Oregon NASA Space Grant AFRIP poster, 2024-2025 project. Authors follow the PDF. Accessed 4 November 2026
[150]	OreSat Project. oresat-simulator @ 2ddcded, 2025. URL https://github.com/oresat/oresat-simulator/tree/2ddcded148263a437108f60b7fe43ac69955acbc. new_sim branch, Basilisk/USB attenuation integration. Accessed 4 November 2026
[151]	AcubeSAT Team. gnuradio_flowgraphs @ master, . URL https://gitlab.com/acubesat/comms/software/gnuradio_flowgraphs/-/tree/master. Accessed 3 November 2026
[152]	OreSat Project. oresat-live-software, . URL https://github.com/oresat/oresat-live-software. Accessed 3 November 2026
[153]	AcubeSAT Team. payload-campaign-ui @ main, . URL https://gitlab.com/acubesat/ops/payload-campaign-ui/-/tree/main. Accessed 3 November 2026
[154]	UniClOGS Project. ultra @ 58e1e32, . URL https://github.com/uniclogs/ultra/tree/58e1e32ebb70356507a762c0424427f893941d5b. Accessed 3 November 2026

Ref.	Source
[155]	FloripaSat Team. received-data, . URL https://github.com/floripasat/received-data. Accessed 3 November 2026
[156]	OreSat Project. oresat-vpn, . URL https://github.com/oresat/oresat-vpn. Accessed 3 November 2026
[157]	Apiwat Jirawattanaphol. Design and Development of Ground Station Network for CubeSats Constellation, Joint Global Multi-Nation Birds, 2017. URL https://kyutech-laplace.net/Alumni/2016/M2/apiwat/M2-apiwat-summary.pdf. Two-page master-course thesis summary. Earlier BIRDS-1 preflight architecture/test. Accessed 4 November 2026
[158]	OreSat Project. oresat-decoders, . URL https://github.com/oresat/oresat-decoders. Accessed 3 November 2026
[159]	Libre Space Foundation. gitlab.com / upsat-plotter, . URL https://gitlab.com/librespacefoundation/upsat/upsat-plotter. Accessed 3 November 2026
[160]	FloripaSat Team. fsat-decoder, . URL https://github.com/floripasat/fsat-decoder. Accessed 3 November 2026
[161]	AcubeSAT Team. archiving-service @ main, . URL https://gitlab.com/acubesat/ops/archiving-service/-/tree/main. Accessed 3 November 2026
[162]	BIRDS-4 Project and Daisuke Nakayama. *BIRDS-4 CW Format and Timing, with CW decoder operator guide*, 2021. URL https://birds4.birds-project.com/document/the-publications-for-amateur-radio-operator. Specification 11 March 2021, guide 14 March 2021. Guide v1.1 and binary download v1.2 differ. Accessed 4 November 2026
[163]	Jacy Chang. BIRDS-RPM ADCS Ground Station: telemetry test dashboard and public manuals @ 1fe3ca3, 2026. URL https://github.com/BIRDSOpenSource/ADCS-TM-Dashboard/tree/1fe3ca31167800157df5ce93ea1d789cf4452d4c. Released packet definitions are generic demonstrations. Accessed 4 November 2026